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Global Beats, Local Tongue

STUDYING CODE SWITCHING
IN K-POP HITS ON BILLBOARD CHARTS

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Outline

- Background and Motivation
- Code Switching vs Code Mixing
- Data collection
- Statistical Test
- Conclusion and Future Direction

Global Beats, Local Tongue: Studying Code Switching in K-pop Hits on Billboard Charts

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Abstract

Code switching, particularly between Korean and English, has become a defining feature of modern K-pop, reflecting both aesthetic choices and global market strategies. This paper is a primary investigation into the linguistic strategies employed in K-pop songs that achieve global chart success, with a focus on the role of code-switching and English lyric usage. A dataset of K-pop songs that appeared on the Billboard Hot 100 and Global 200 charts from 2017 to 2025, spanning 14 groups and 8 solo artists, was compiled. Using this dataset, the proportion of English and Korean lyrics, the frequency of code-switching, and other stylistic features were analysed. It was found that English dominates the linguistic landscape of globally charting K-pop songs, with both male and female performers exhibiting high degrees of code-switching and English usage. Statistical tests indicated no significant gender-based differences, although female solo artists tend to favour English more consistently. A classification task was also performed to predict performer gender from lyrics, achieving macro F1 scores up to 0.76 using multilingual embeddings and handcrafted features. Finally, differences between songs charting on the Hot 100 versus the Global 200 were examined, suggesting that, while there is no significant gender difference in English, higher English usage may be more critical for success in the US-focused Hot 100. The findings highlight how linguistic choices in K-pop lyrics are shaped by global market pressures and reveal stylistic patterns that reflect performer identity and chart context.

Keywords: Code Switching, Computational Linguistics, Korean Pop, K-Pop, Statistical Analysis

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BACKGROUND AND MOTIVATION



As 'Golden' Remains at No. 1, 'KPop Demon Hunters' Now Has Four Songs in the Hot 100's Top 10 — a Historic First for a Soundtrack

By Chris Willman



Billboard launches K-pop music charts for first time to track genre's local and global metrics



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KPop Demon Hunters Earns Three Golden Globe Nominations



Huntr/X don't miss. KPop [Demon Hunters](#), directed by [Maggie Kang](#) and [Chris Appelhans](#), continues its global domination that started with the film's launch on [Netflix](#) in June 2025, and hasn't stopped going up, up, up since. The animated musical epic has nabbed three nominations for the 83rd annual Golden Globes: Best Animated Motion Picture, Best Cinematic and Box Office Achievement, and Best Original Song from a Motion Picture for "Golden," with music by [Joong Gyu Kwak](#), [Yu Han Lee](#), [Hee Dong Nam](#) (Ido), [Jeong Hoon Seo](#) (24), [Park Hong Jun](#) (Teddy); and lyrics by [Ejae](#) and [Mark Sonnenblick](#).

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BTS EARNS FIRST NO. 1 ALBUM ON BILLBOARD 200 CHART WITH 'LOVE YOURSELF: TEAR'

Korean boy band BTS debuts at No. 1 on the Billboard 200 albums chart with its latest effort, "Love Yourself: Tear." It's the first No. 1 for the seven-member group, and the first K-pop album to lead...

By **Keith Caulfield** ▾

BLACKPINK'S MAJOR DEBUT: NEW K-POP GIRL GROUP LANDS NO. 1 & 2 ON WORLD DIGITAL SONGS CHART

Blackpink is less than a month old, but the new girl group is already making major moves on the Billboard charts with their debut singles "Boombayah" and "Whistle."

By **Jeff Benjamin** ▾

Global Rise of K-pop

Rapid expansion
into Western
markets (Hot 100,
Global 200).

Increasing visibility
of multilingual
lyrics (Korean +
English).

English used for
“global vibe,” fun
indexing, fan
connection.

Music videos +
sonic
experimentation →
global appeal.

Why Study Code-Switching in K-pop?

- English–Korean code switching has become a norm, not an exception.
 - *Frequent code-switch events (≈ 17 –25 per song depending on groups/solo acts)*
- Lack of **computational** analysis of this phenomenon.
- **Research questions:**
 - *Do male vs. female performers differ in English/Korean usage?*
 - *Does code-switching differ by gender or chart type?*
 - *Can performer gender be predicted purely from lyrics?*

CODE SWITCHING VS CODE MIXING

Definitions



Code-Switching (CS)

Alternating between languages at boundaries (line, phrase, clause).



Code-Mixing

Blending languages within phrases/words.



K-pop lyrics typically show:

Inter-sentential switching (“Korean verse → English pre-chorus”)

Intra-sentential switching (“Hangul + English words together”)

Tag-switching (“baby,” “yeah,” “okay,” “let’s go”)

Example

미쳐 that candy's sassy

michyeo

ko-ko

en-en

뉴런의 체계를 갱신

nyuleon-ui

chegyeleul

gaengsin

ko-ko

Why CS Matters for K-pop



USED FOR RHYTHM,
EMPHASIS, HOOKS.



ENGLISH OFTEN MARKS
“GLOBAL-NESS.”



CAN INFLUENCE CHART
PERFORMANCE AND
CULTURAL INDEXING.

DATA COLLECTION

Artist & Song Selection



Time period: Jan 2017 –
May 2025.



Sources:

*Lyrics → Genius (via LyricsGenius).
Chart data → Billboard Hot 100 +
Global 200.*



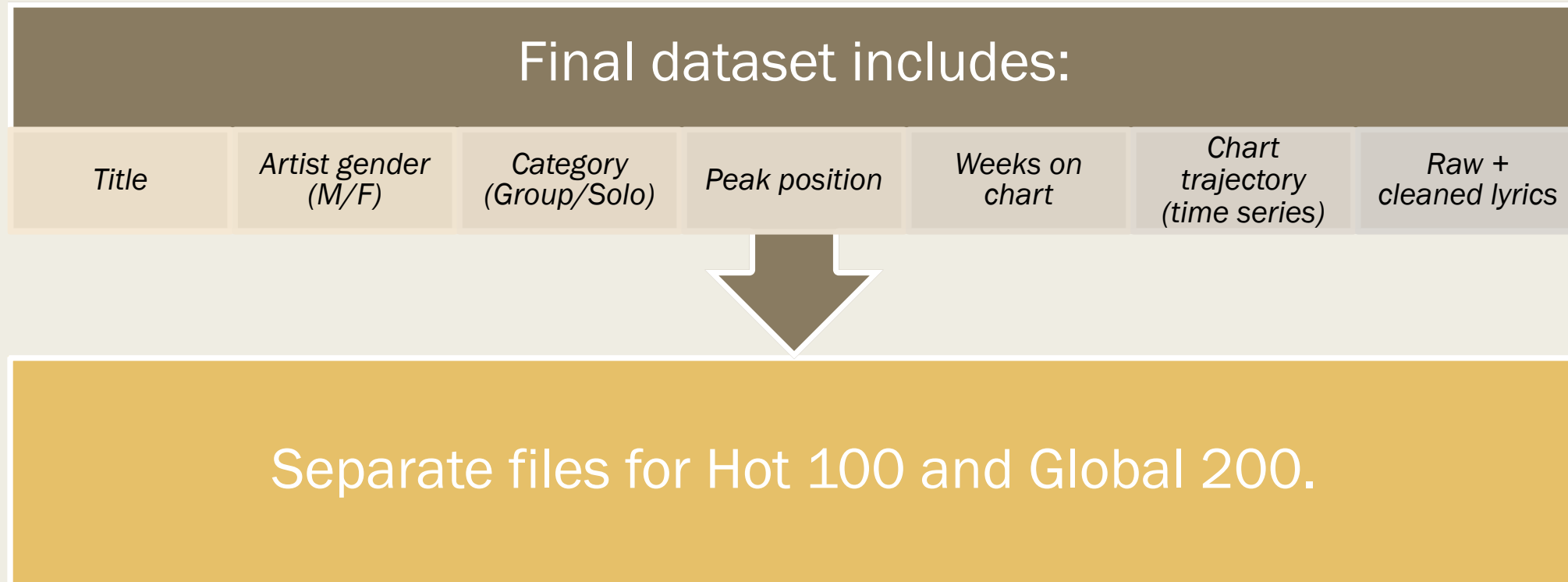
22 artists total:

*Groups: 7 female, 4 male
Soloists: 4 female, 6 male*



All songs the artists had on
either chart (hot 100 or
global 200) were retrieved.

Dataset Construction



Dataset Stats



Most represented:

BTS (29 songs), BLACKPINK (24).



Soloists:

V (15), Jungkook (14), Jennie (14).



Nearly all artists appear on
Global 200;

STATISTICAL TESTS

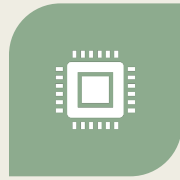
Feature Engineering for classification



KOREAN LYRIC RATIO



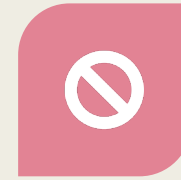
ENGLISH LYRIC RATIO



CODE-SWITCH COUNT



REPETITION COUNT



ALLITERATION



FILLER WORDS ("OH,"
"YEAH," "LA," ETC.)



COMBINED WITH
MULTILINGUAL
EMBEDDINGS FOR
THE CLASSIFICATION
TASK.

Statistical Methods

Normality check →
Shapiro-Wilk

This revealed that
most features were
non-normal.

Welch's t-test

This test showed us
unequal variance.

Proceeded with a
Mann-Whitney U
for a nonparametric
test.

Set the
Significance
threshold to $\alpha =$
0.05.

Key Hypotheses

1. Boy vs. Girl Groups → English/Korean ratio differences?
2. Gender differences in number of code-switches?
3. Soloists vs groups?
4. Hot 100 vs Global 200 linguistic differences?

FINDINGS

Findings — Classification

- Boy vs girl groups:
 - *Hot 100*: 76% accuracy and a macro F1-score of 0.76
 - *Global 200*: 72% accuracy and a macro F1-score of 0.71.
- Conclusion
 - Although individual linguistic features (such as the Korean lyric ratio) did not yield significant differences across genders, their combination with learned representations allowed models to predict performer gender at a level significantly above chance.

Findings — Code-Switching Frequency

- Boy vs girl groups:
 - *Hot 100: 19.05 vs 16.96 switches → NOT significant.*
 - *Global 200: 20.57 vs 25.32 → NOT significant.*
- Conclusion: High code-switching across all categories; gender doesn't matter.

Findings — Korean vs English Ratio

- Trend:
 - *Boy groups use **slightly** more Korean.*
 - *Girl groups use **slightly** more English.*
- But Mann–Whitney: NOT statistically significant, therefore we need more data and variety of groups to study on.
- Female soloists, however, are far more English-heavy in their songs (generally >95% in some tracks).

Chart Context Effects

- Boy groups: No difference Hot100 vs G200 → stable Korean ratio.
- Girl groups: Slight trend toward more English on Hot 100 (however not statistically significant).
 - *Hot 100 is a US-centric chart.*
 - *This could suggest a strategic push for English usage for U.S.-oriented releases.*

Interpretation

English dominance
consistent across
gender and chart.

Trends exist but lack
statistical power
due to less data and
limited time period.

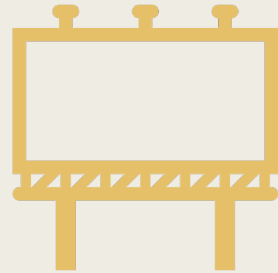
Strong stylistic
similarity across
groups → English as
a global identity
marker.

CONCLUSION AND FUTURE DIRECTIONS

Main Takeaways

- English has become a dominant linguistic component of global K-pop.
- Both male and female acts code-switch heavily and frequently.
- Gender differences in language ratio exist but are not statistically significant.
- Classification models achieve up to 76% macro F1, showing latent stylistic cues exist beyond raw language ratios.

Contributions



**First computational analysis
linking:**

Code-switching

Billboard chart context



**Novel dataset combining chart
metadata + multilingual lyrics.**

Limitations



Only internationally
successful songs →
bias.



Small sample size.



No audio/visual context
(important for K-pop's
multimodality).



Simplifies linguistic
structure of Korean.

Future Work



Expand dataset (domestic charts, pre-2017, smaller agencies).



Include multimodal features (audio, choreography, MV aesthetics).



Chart-success prediction using linguistic + non-linguistic features.



Collaboration with Korean linguists for deeper semantic analysis.



Temporal analysis: English rise over time.



Listener reception analysis across markets.

THANK YOU

Send me an email or dog pictures or playlists or job offers :)

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<https://callmesanfornow.github.io>